



SemAlign: Language Guided Semi-Supervised Domain Generalization

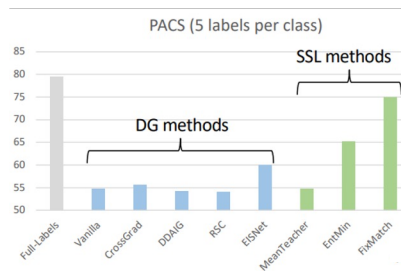
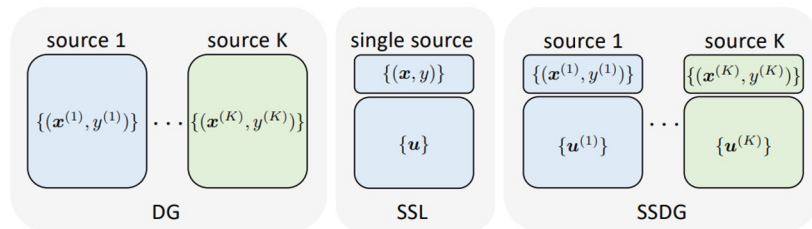
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University of Moratuwa

Session O2-B.T2: Vision And Language

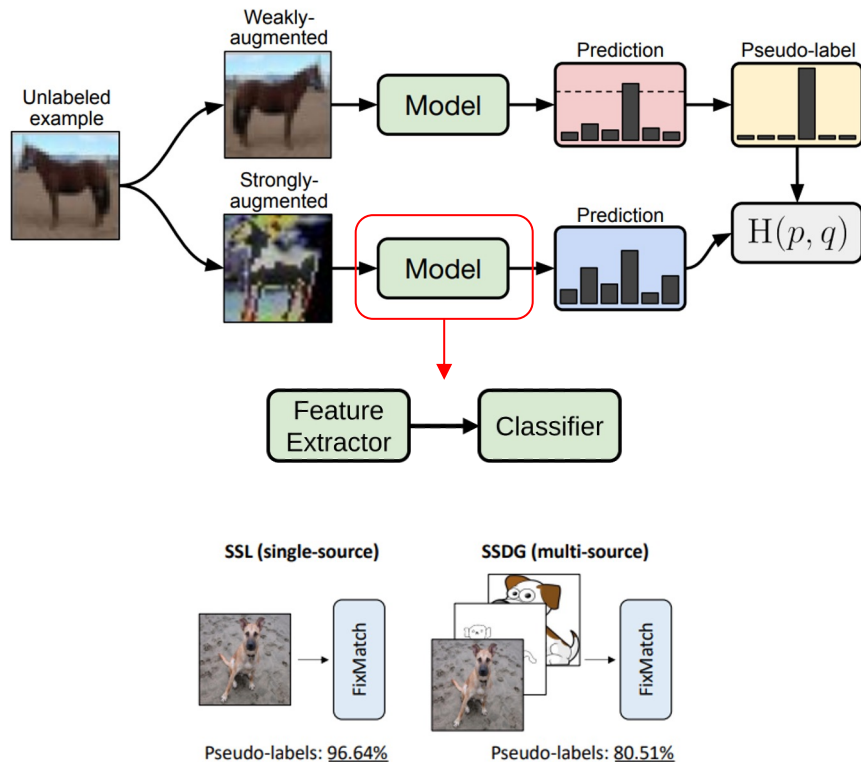
Semi Supervised Domain Generalization

- Computer vision models should be
 - Generalizable for unseen domain shifts
 - Data efficient
- Domain Generalization (DG)
 - Learn models robust against domain shifts
- Semi Supervised Learning (SSL)
 - Learn models with limited labels
- Semi Supervised Domain Generalization (SSDG)
 - DG methods struggle in SSDG settings
 - SSL methods perform better



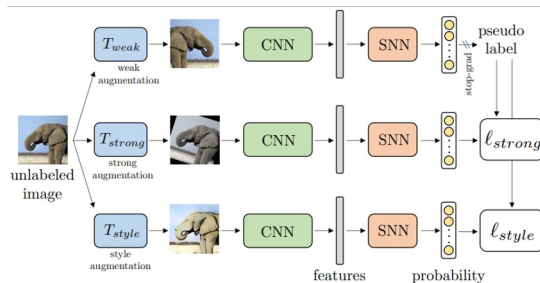
FixMatch

- FixMatch is a SSL training method which combines,
 - Consistency regularization
 - Pseudo Labeling
- Works well for single domain, but struggles in the presence of domain shifts
- Challenges
 - Accurate Pseudo Labels (PL)
 - Overfitting to limited labels
 - Number of samples above the confidence threshold (effective data utilization)
 - Low diversity of PLs

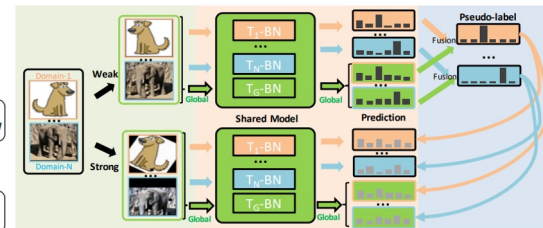


Existing Methods

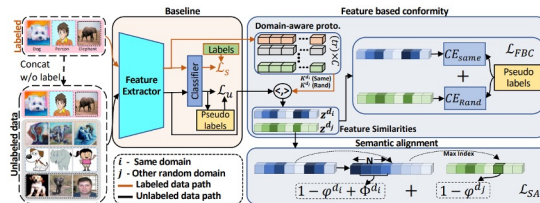
- Input level constraints
 - Style augmentations
- Feature level constraints
 - Domain feature alignment
- Output level constraints
 - Stochastic classifiers
 - Domain specific classifiers.
 - Weight modulation for domain specific feature learning.



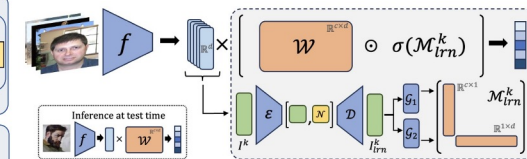
StyleMatch (NeurIPS-W 2021)



MultiMatch (ACM TOMM 2024)



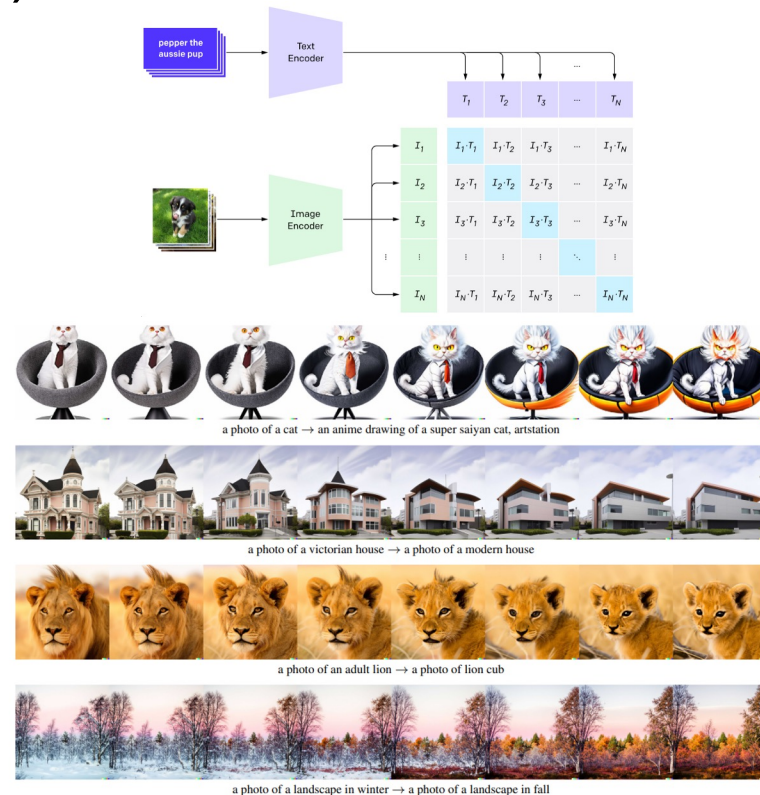
FBCSA (CVPR 2024)



DGWM (WACV 2025)

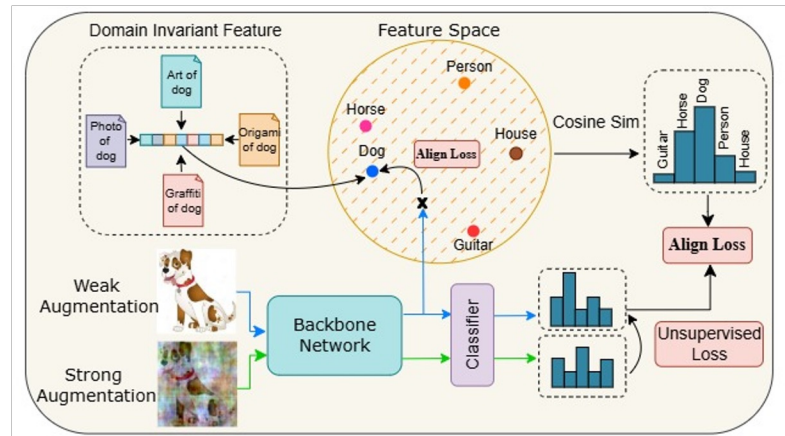
Vision Language Models (VLMs)

- Generalized features from large-scale vision-language pretraining.
- Shared image-text embedding space.
- Algebraic operations on text embeddings can induce corresponding semantic changes in the visual domain.
- This suggests that textual representations sufficiently encode the information for visual composition.



Contributions

- We study the SSDG problem and propose a novel approach from three perspectives,
 - Pixel space
 - Feature space
 - Model output space
- We leverage domain invariant class priors from the rich embedding space of a VLM to drive domain invariant visual representation learning.
- We highlight the importance of considering PL accuracy, effective data utilization (EDU) and PL diversity in SSDG settings.

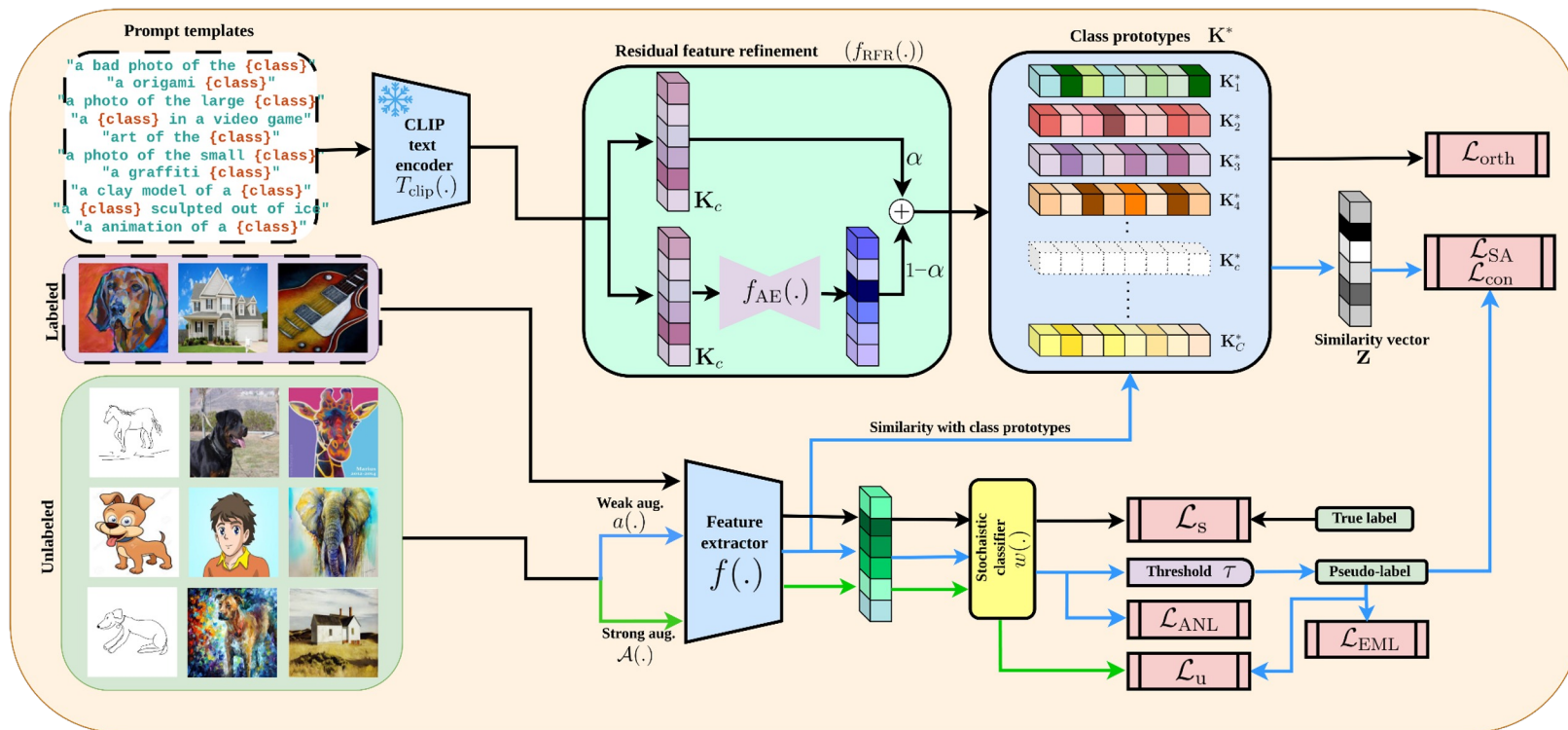


Experiments

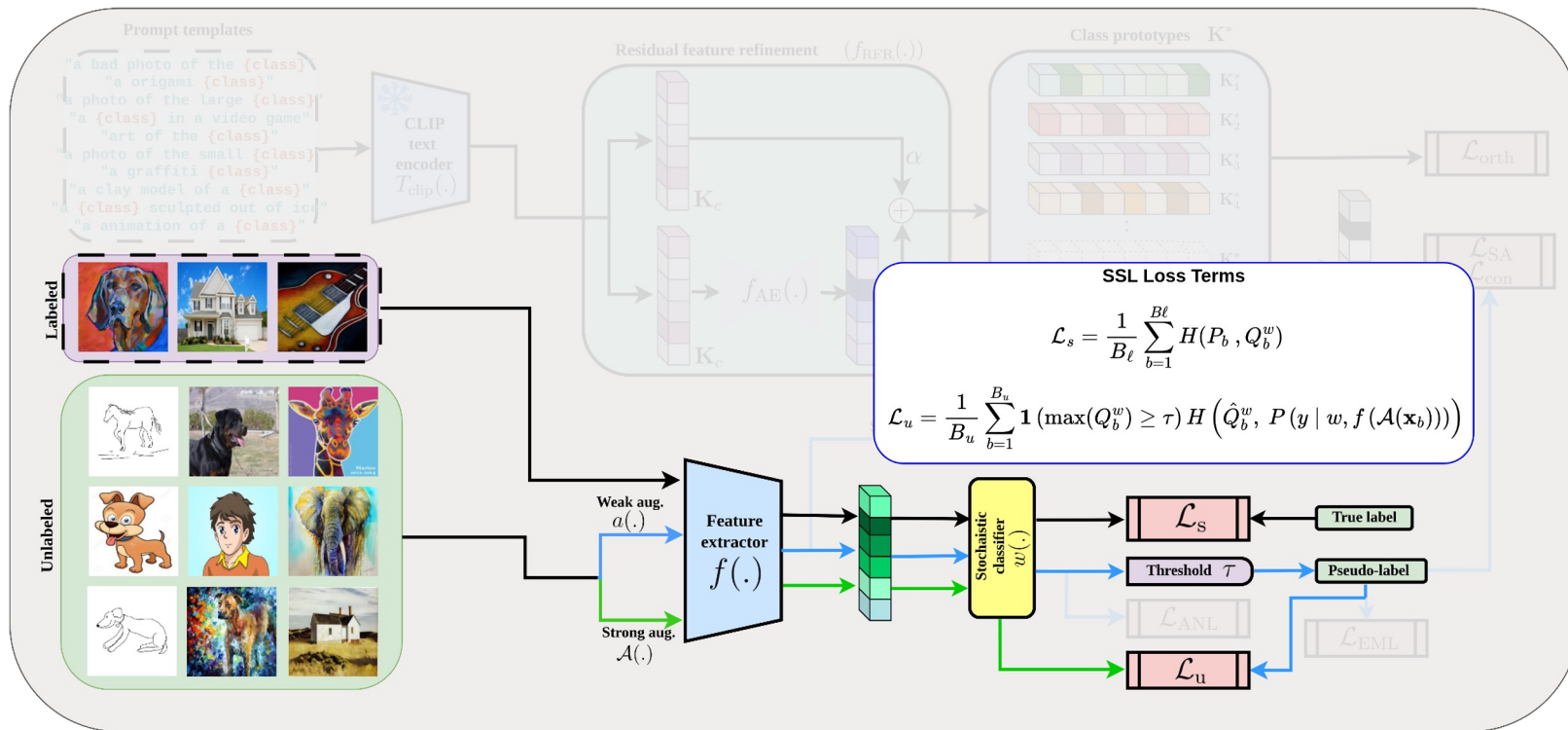
- Experiments are conducted on 4 benchmark datasets.
- Leave one domain out protocol for training and testing.
- Limited labeled samples per class for training domains.
 - 5 and 10 labels



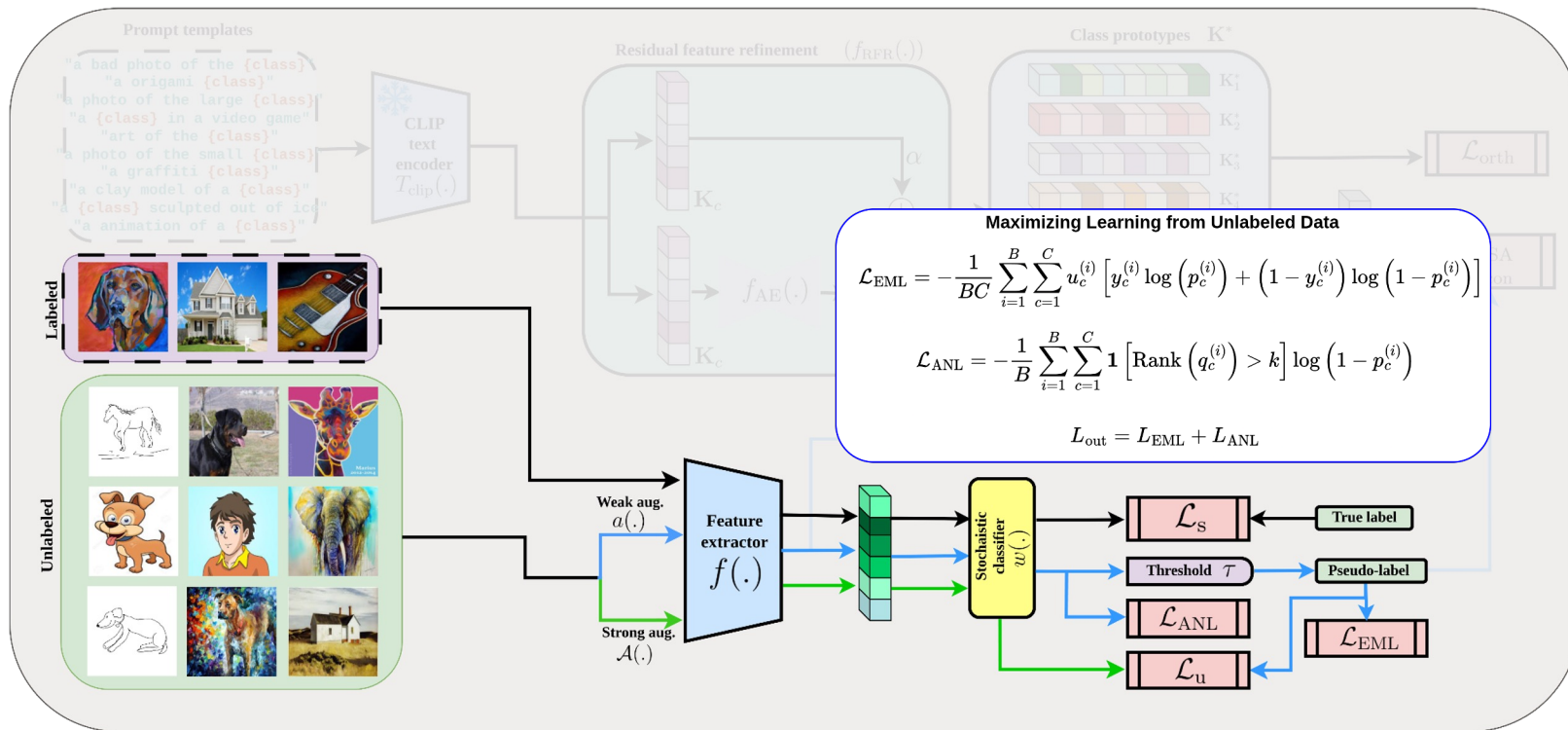
Methodology



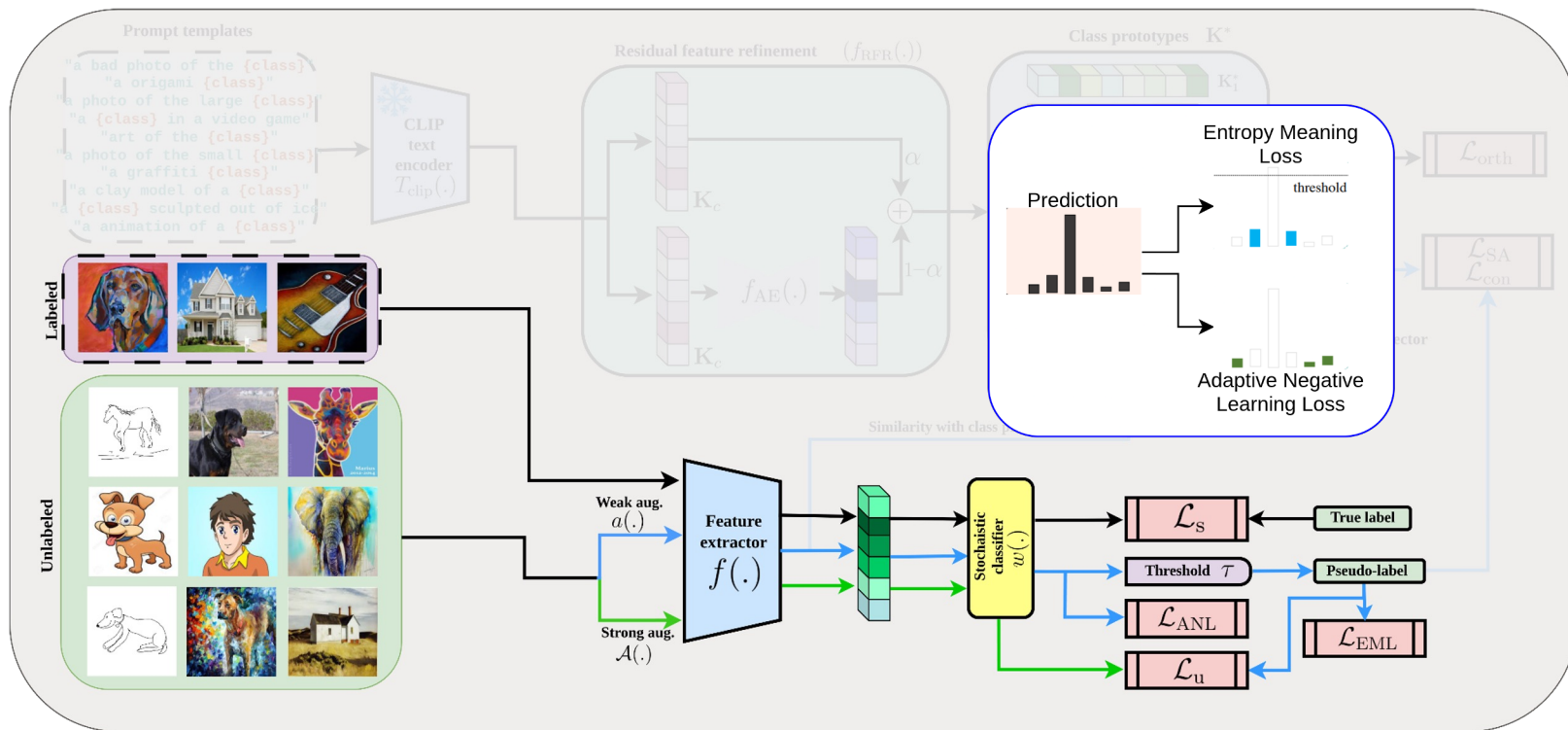
Methodology



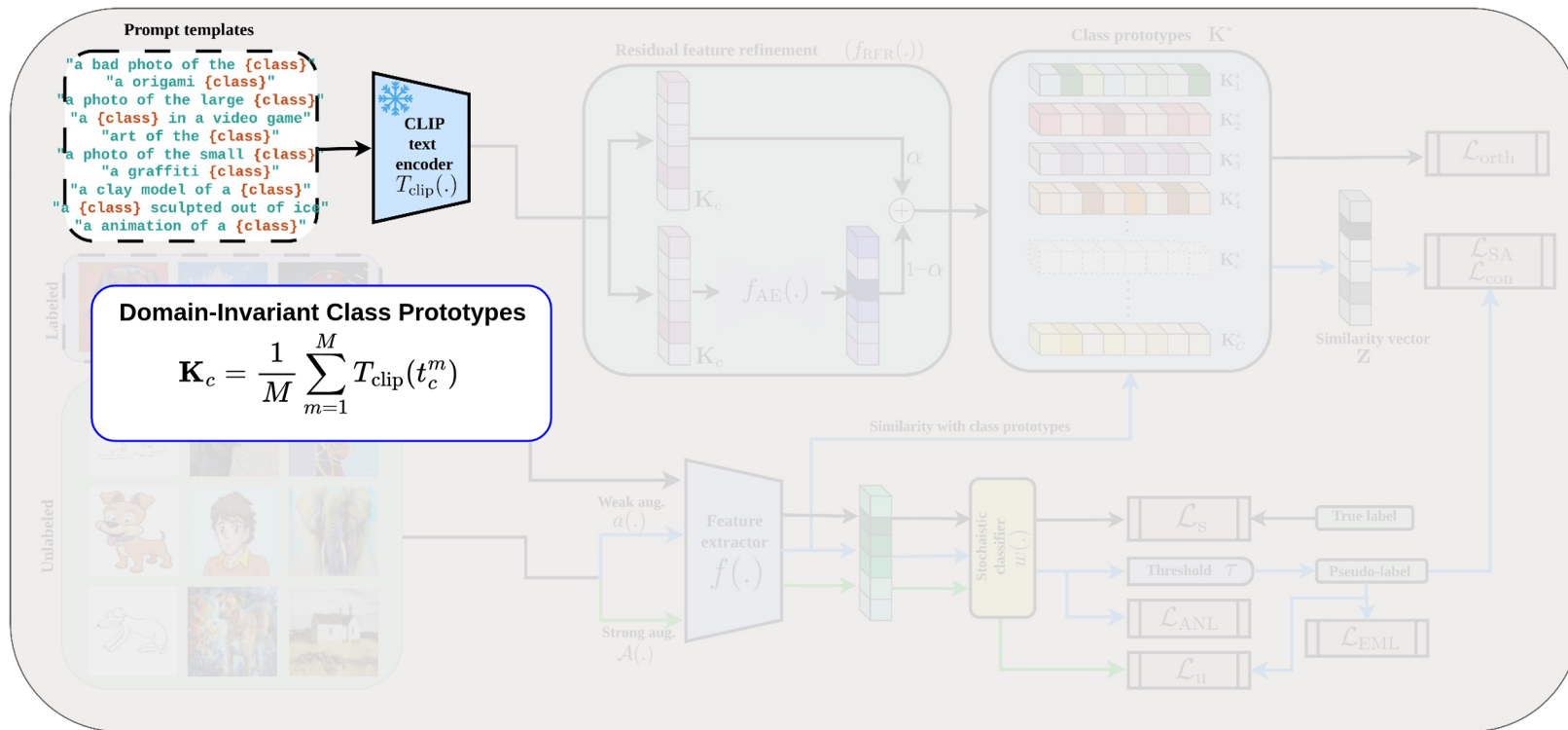
Methodology



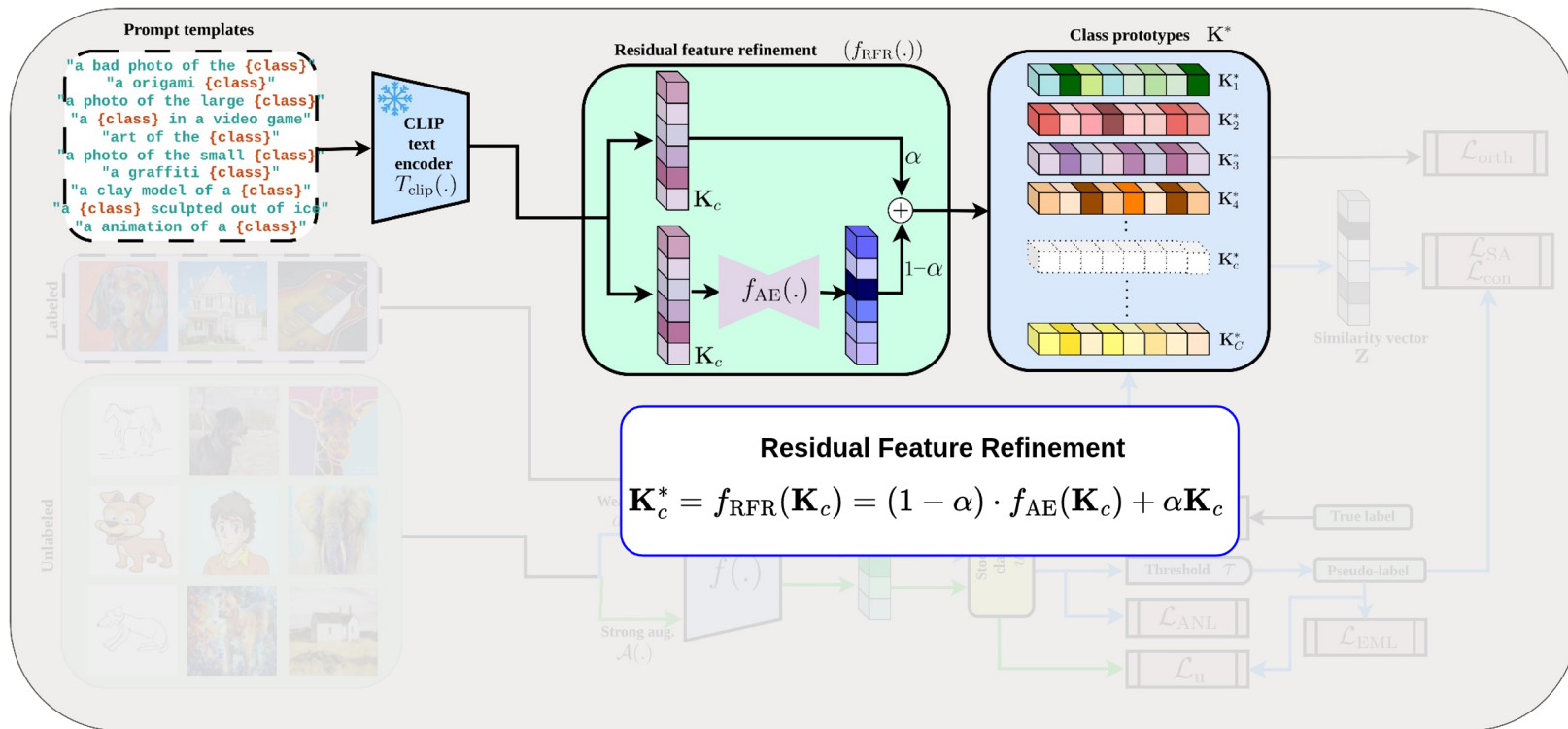
Methodology



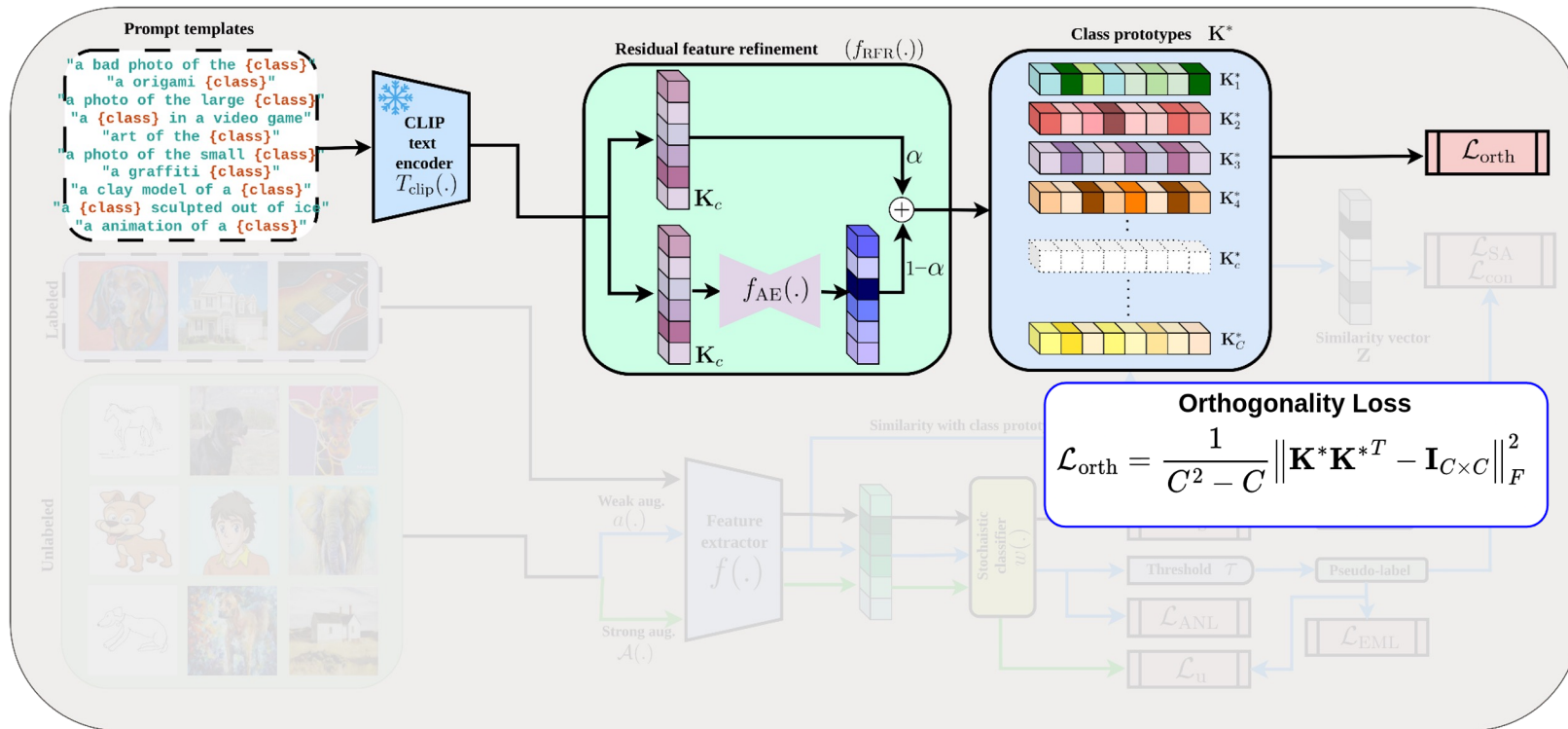
Methodology



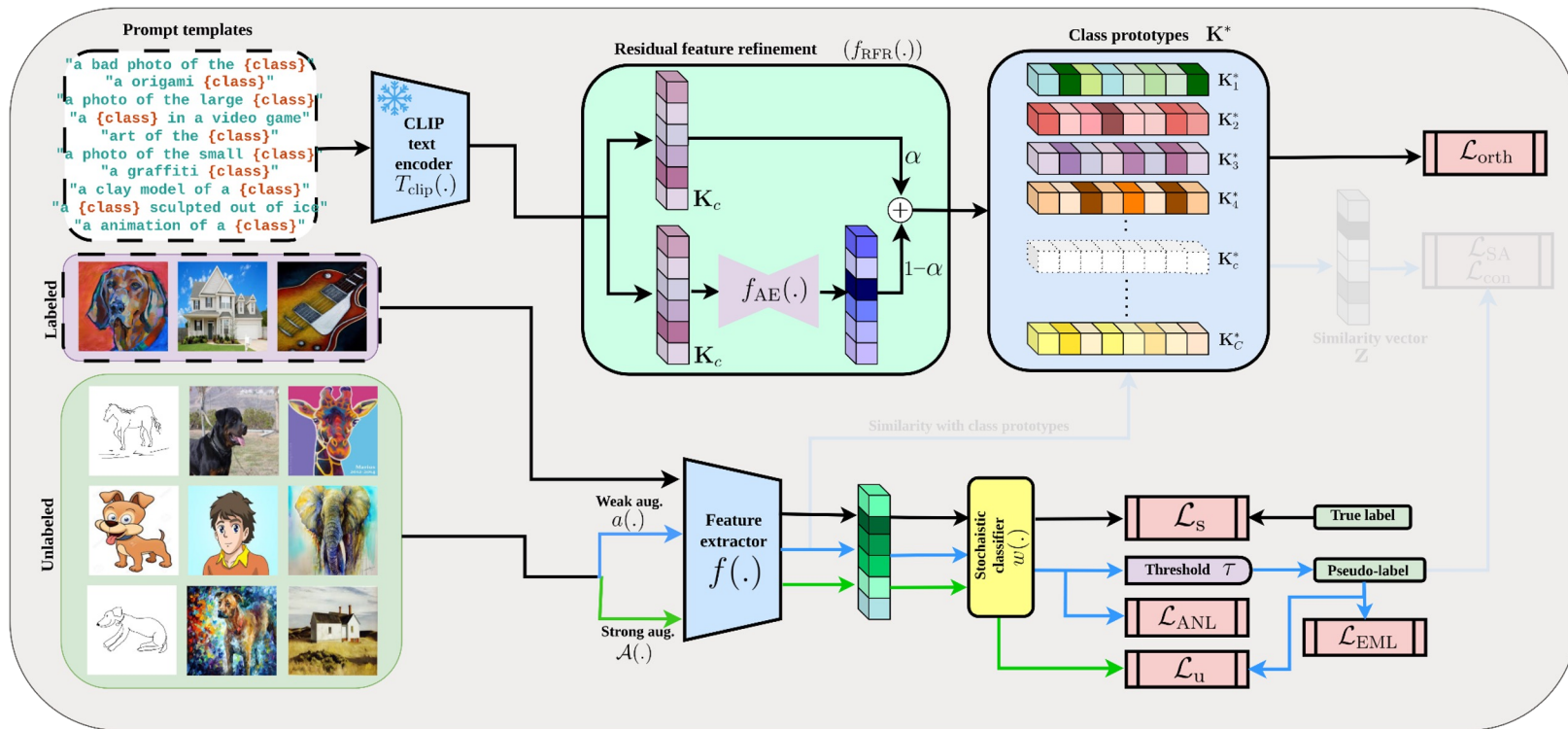
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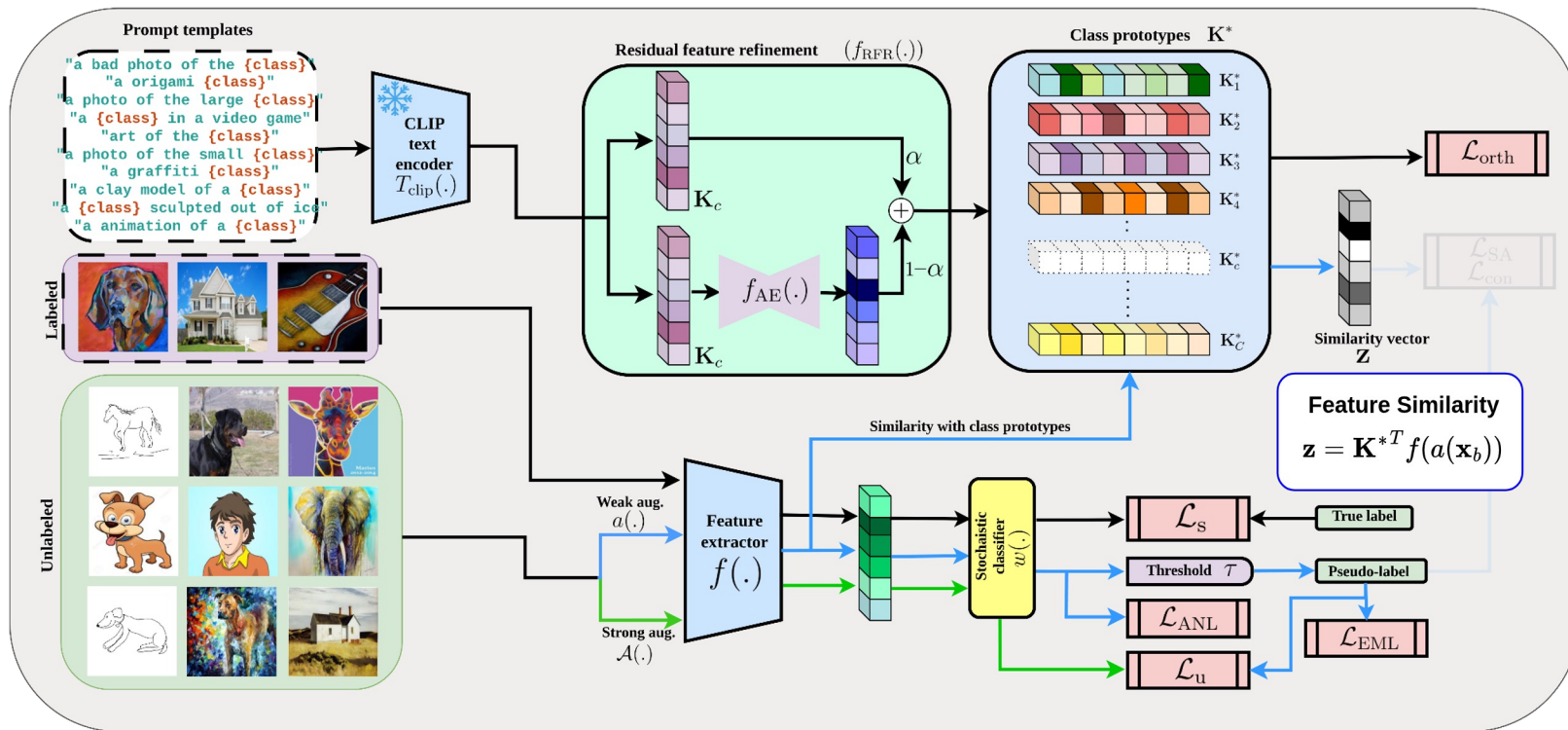
Methodology



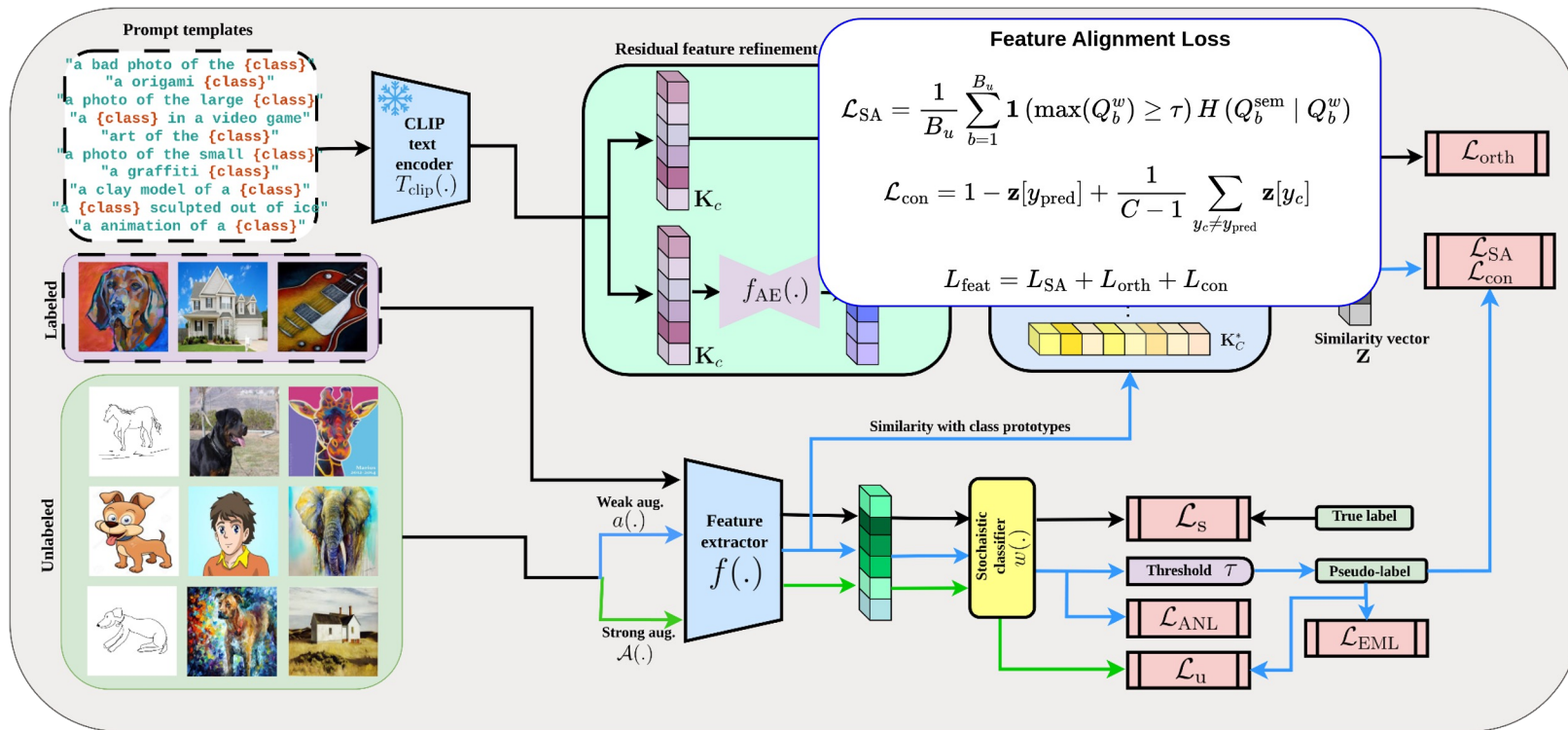
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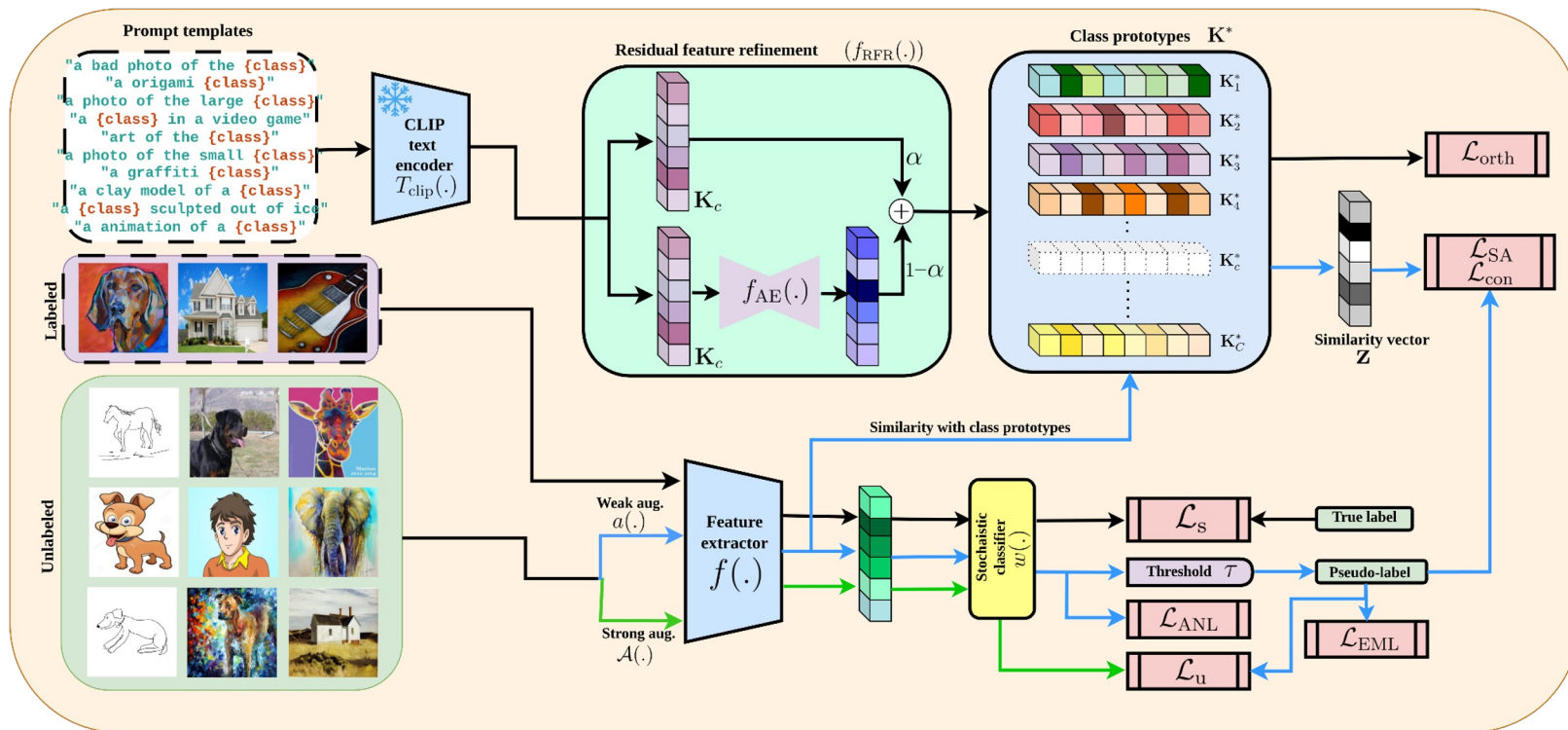
Methodology



Methodology



Methodology



Results

# labeled data Model	5 labels				
	PACS	OfficeHome	DigitsDG	miniDomainNet	Avg
FixMatch [63]	75.2 ± 1.3	55.2 ± 0.9	55.0 ± 2.8	34.6 ± 1.3	55.0
StyleMatch [94]	<u>78.4 ± 0.9</u>	<u>56.3 ± 0.6</u>	55.9 ± 1.6	<u>36.6 ± 1.0</u>	56.8
FBC-SA [20]	75.4 ± 1.1	55.8 ± 0.7	<u>64.3 ± 0.5</u>	36.1 ± 1.2	<u>57.9</u>
DGWM [21]	78.0 ± 0.8	<u>56.3 ± 0.7</u>	55.9 ± 0.6	36.4 ± 1.4	56.7
SemAlign (Ours)	79.6 ± 0.5	57.0 ± 0.8	66.3 ± 0.6	36.7 ± 1.2	59.9

# labeled data Model	10 labels				
	PACS	OfficeHome	DigitsDG	miniDomainNet	Avg
FixMatch [63]	77.1 ± 2.0	58.2 ± 1.0	64.5 ± 1.4	39.8 ± 0.6	59.9
StyleMatch [94]	<u>79.7 ± 1.9</u>	<u>59.8 ± 0.7</u>	66.1 ± 1.6	40.6 ± 1.2	61.6
FBC-SA [20]	78.1 ± 2.4	59.3 ± 1.0	<u>69.4 ± 1.4</u>	<u>41.3 ± 1.1</u>	<u>62.0</u>
DGWM [21]	78.8 ± 1.6	59.6 ± 1.0	65.6 ± 1.5	<u>41.3 ± 1.5</u>	61.3
SemAlign (Ours)	81.8 ± 1.7	60.3 ± 0.9	74.2 ± 1.4	42.2 ± 1.3	64.6

We achieve an average performance gain of (**+2%**) and (**+2.6%**) in the 5 and 10 labels cases over previous SOTA.

Pseudo Labels

- Pseudo Labeling Accuracy

$$\text{Pseudo Labeling Accuracy} = \frac{\text{Correct Pseudo Labels}}{\text{Total Pseudo Labels}} \times 100\%$$

- Effective Data Utilization

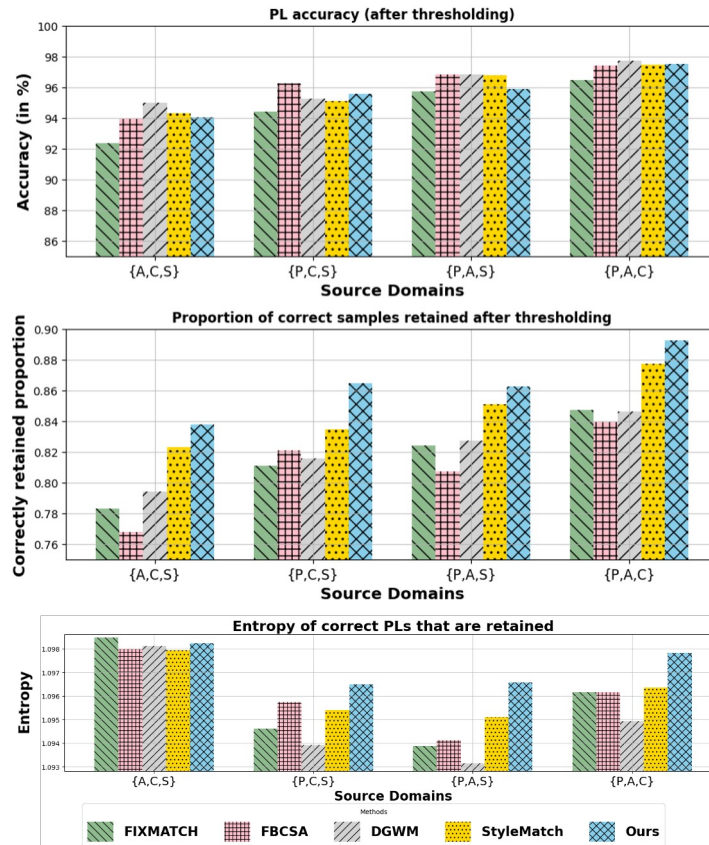
$$\text{Effective Data Utilization} = \frac{\text{Correct Pseudo Labels}}{\text{Total Dataset Size}} \times 100\%$$

- Pseudo Label Diversity

$$\text{Pseudo Label Diversity} = - \sum_{i=1}^D \frac{n_i}{\sum_{j=1}^D n_j} \log \left(\frac{n_i}{\sum_{j=1}^D n_j} \right)$$

n_i = Number of pseudo labels from source domain i

D = Number of source domains



Ablations

Contribution of components,

- Proposed data augmentation scheme gives a 1.2% gain.
- Language guidance gives a 2.3% gain.

Scaling Performance with Number of labels per class

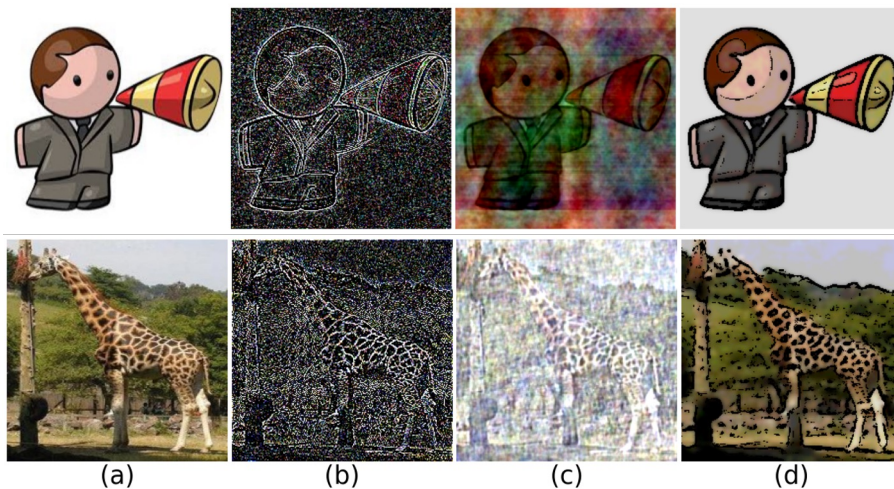
- Fully supervised ERM baseline which achieves 80.0 ± 0.5 .
- With only 10 labels per-class labels, our method surpasses the fully supervised baseline

Method	Average
Baseline [63]	77.1
Baseline + SC	78.1
Baseline + SC + $\mathcal{L}_{out}$	78.3
Baseline + SC + $\mathcal{L}_{out}$ + Data aug	79.5
Baseline + SC + $\mathcal{L}_{out}$ + Data aug + $\mathcal{L}_{SA}$	80.8
Baseline + SC + $\mathcal{L}_{out}$ + Data aug + $\mathcal{L}_{SA}$ + $\mathcal{L}_{con}$	81.1
Baseline + SC + $\mathcal{L}_{out}$ + Data aug + $\mathcal{L}_{SA}$ + $\mathcal{L}_{orth}$	81.2
Baseline + SC + Data aug + $\mathcal{L}_{SA}$ + $\mathcal{L}_{orth}$ + $\mathcal{L}_{con}$	81.0
Baseline + SC + $\mathcal{L}_{out}$ + Data aug + $\mathcal{L}_{SA}$ + $\mathcal{L}_{orth}$ + $\mathcal{L}_{con}$ (Ours)	81.8

Model	5	10	25	50
ERM [49]	51.2 ± 1.0	59.8 ± 2.5	66.7 ± 2.2	71.2 ± 1.9
FixMatch [51]	75.2 ± 1.3	77.1 ± 2.0	77.6 ± 1.4	78.7 ± 1.5
SemAlign (Ours)	79.6 ± 0.5	81.8 ± 1.7	82.2 ± 1.2	83.3 ± 1.2

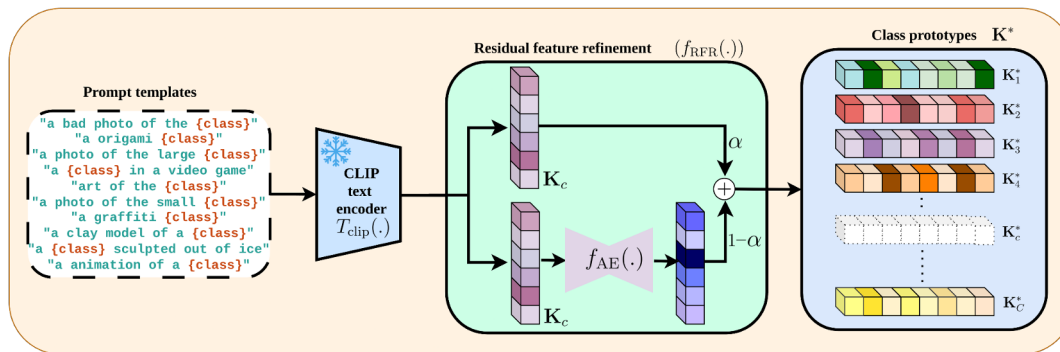
Data Augmentations

- Fourier based augmentations
 - Constant amplitude component
 - Swapping amplitudes using a dynamically updated amplitude bank
- Texture reduction
- No additional models are used
- Less overhead

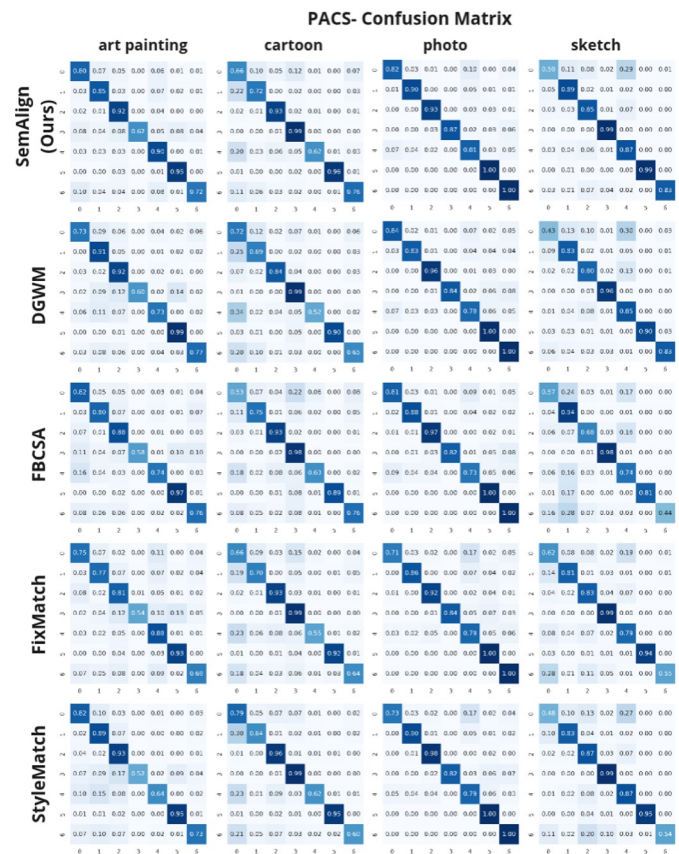
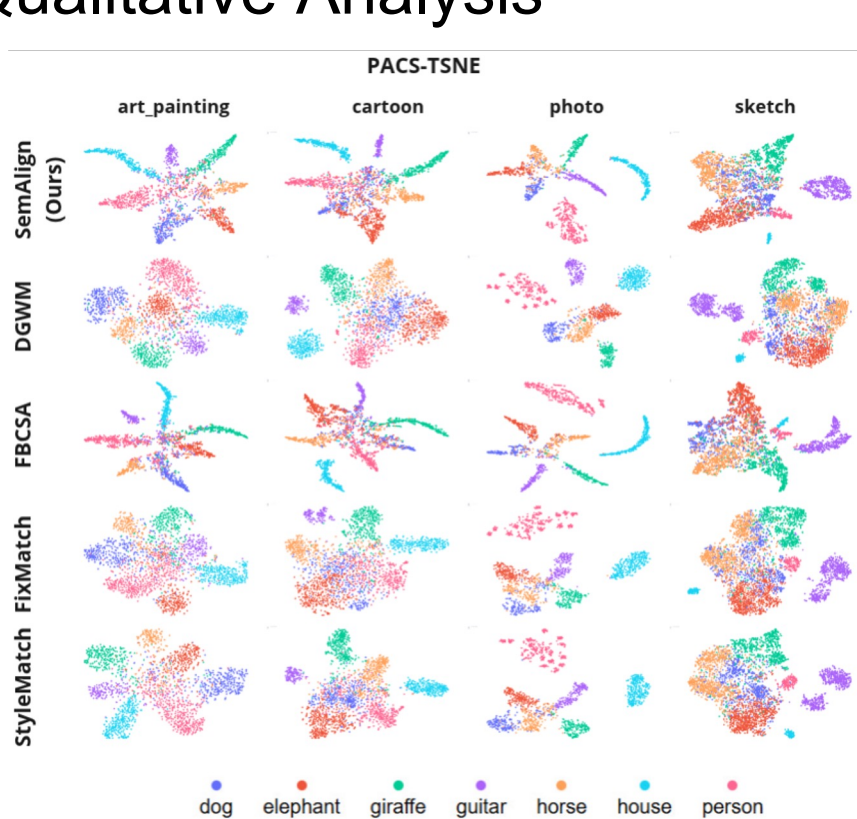


Method	Average time/epoch (sec)	Overhead
FixMatch [63]	38.53	-
StyleMatch [94]	97.54	153.15%
SemAlign (Ours)	60.2	56.24%

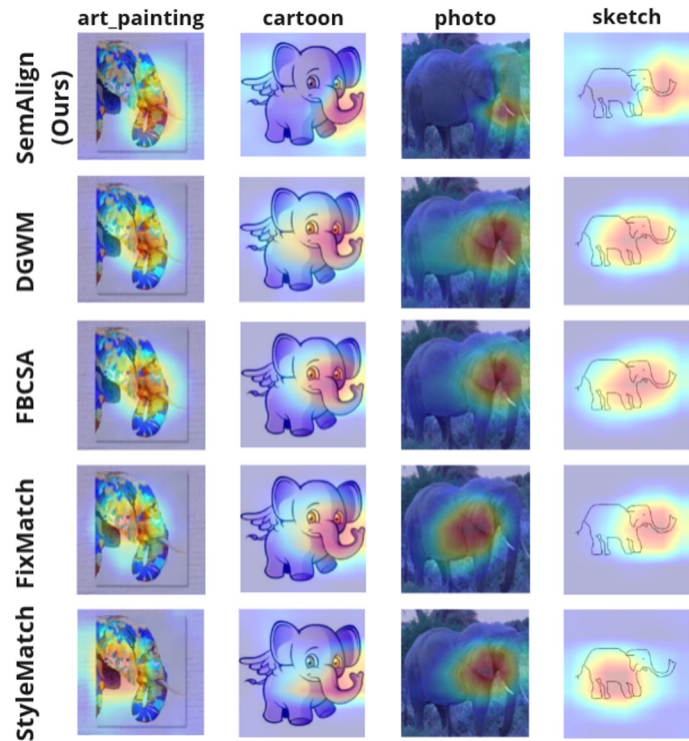
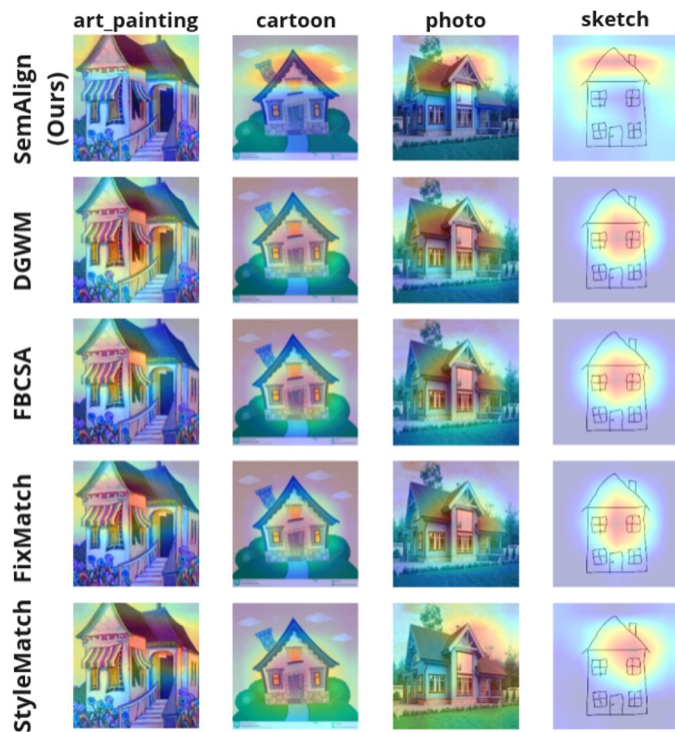
Residual Feature Refinement



Qualitative Analysis



Qualitative Analysis



Conclusion

- A novel approach for semi-supervised domain generalization, addressing the problem from three complementary directions,
 - Input space
 - Feature space
 - Model output space
- Leverages class semantics to encourage discriminative, domain-invariant feature learning.
- Achieves higher effective utilization of unlabeled data with more diverse pseudo-labels.
- Show consistent and notable gains over existing SSDG methods over four datasets.

Thanks for watching



Github page

Session O2-B.T2: Vision And Language